

A. Instruction Set Reference

Opcode	Mnemo	Operands	Operation
00000	R-Type	rs rt rd aux	see table A.2
00010	J	address	PC = JumpAddr ⁽¹⁾
00011	JAL	address	R[31] = PC + 4; PC = JumpAddr
00100	ANDI	rs rt imm	R[rt] = R[rs] & ZeroExtImm ⁽²⁾
	ANDIH	rs rt imm	R[rt] = R[rs] & ZeroCatImm ⁽³⁾
00101	ORI	rs rt imm	R[rt] = R[rs] ZeroExtImm
	ORIH	rs rt imm	R[rt] = R[rs] ZeroCatImm
00100	XORI	rs rt imm	R[rt] = R[rs] $\oplus$ ZeroExtImm
	XORIH	rs rt imm	R[rt] = R[rs] $\oplus$ ZeroCatImm
01000	LW	rs rt imm	R[rt] = M[addr] ⁽⁴⁾
01001	SW	rs rt imm	M[addr] = R[rt]
01010	SH	rs rt imm	M[addr](15:0) = R[rt](15:0)
01011	SB	rs rt imm	M[addr](7:0) = R[rt](7:0)
01100	LH	rs rt imm	R[rt] = {16{M[addr](15:0)[15], M[addr](15:0)} ⁽⁵⁾
01101	LHU	rs rt imm	R[rt] = {16'b0, M[addr](15:0)}
01110	LB	rs rt imm	R[rt] = {24{M[addr](7:0)[7]}, M[addr](7:0)} ⁽⁶⁾
01111	LBU	rs rt imm	R[rt] = {24'b0, M[addr](7:0)}
10000	BEQ	rs rt imm	if (R[rs] == R[rt]) PC=PC+4+BranchAddr ⁽⁷⁾
10001	BNE	rs rt imm	if (R[rs] != R[rt]) PC=PC+4+BranchAddr
10010	BLT	rs rt imm	if (R[rs] < R[rt]) PC=PC+4+BranchAddr
10011	BGT	rs rt imm	if (R[rs] > R[rt]) PC=PC+4+BranchAddr
10100	BLE	rs rt imm	if (R[rs] <= R[rt]) PC=PC+4+BranchAddr
10101	BGE	rs rt imm	if (R[rs] >= R[rt]) PC=PC+4+BranchAddr
10110	SLTI	rs rt imm	R[rt] = (R[rs] < SignExtImm) ? 1 : 0
10111	SLTIU	rs rt imm	R[rt] = (R[rs] < SignExtImm) ? 1 : 0 ⁽⁸⁾
11000	ADDI	rs rt imm	R[rt] = R[rs] + SignExtImm
11000	LUI	rt imm	R[rt] = ZeroCatImm

Tabla A.1: RTM32 Instruction Set sorted by opcode

- (1) JumpAddr = {PC+4[31:29], address, 2'b0}
- (2) ZeroExtImm = {16{1b'0}, imm}
- (3) ZeroCatImm = {imm, 16{1b'0}}
- (4) addr = R[rs] + {15{imm[16]}, imm}
- (5) Extend halfword to word
- (6) Extend byte to word
- (7) BranchAddr = {13{imm[16]}, imm, 2'b0}
- (8) Operands considered unsigned numbers (vs. 2's comp.)

Funct	Mnemo	Operands	Operation
000000	SLL	rs rt rd aux	$R[rd] = R[rt] \ll \text{aux}$
000001	SRL	rs rt rd aux	$R[rd] = R[rt] \gg \text{aux}$
000010	SRA	rs rt rd aux	$R[rd] = R[rs] \ggg \text{aux}$
000011	SLLR	rs rt rd	$R[rd] = R[rt] \ll R[rs][4:0]$
000100	SRLR	rs rt rd	$R[rd] = R[rt] \gg R[rs][4:0]$
000101	SRAR	rs rt rd	$R[rd] = R[rt] \ggg R[rs][4:0]$
000110	CFS	rs aux	not implemented yet
000111	CTS	rs aux	not implemented yet
001000	AND	rs rt rd	$R[rd] = R[rs] \& R[rt]$
001001	OR	rs rt rd	$R[rd] = R[rs] \mid R[rt]$
001010	XOR	rs rt rd	$R[rd] = R[rs] \oplus R[rt]$
001011	NOR	rs rt rd	$R[rd] = \neg (R[rs] \mid R[rt])$
001100	SLT	rs rt rd	$R[rd] = (R[rs] < R[rt]) ? 1 : 0$
001101	SLTU	rs rt rd	$R[rd] = (R[rs] < R[rt]) ? 1 : 0^{(8)}$
001110	JR	rs	$PC = R[rs]$
001111	JALR	rs rt	$R[31] = PC + 4; PC = R[rs]$
010000	LHX	rs rt rd	$R[rt] = \{16\{M[R[rs] + R[rd]](15:0)[15], M[R[rs] + R[rd]](15:0)\}$
010001	LHUX	rs rt rd	$R[rt] = \{16'b0, M[R[rs] + R[rd]](15:0)\}$
010010	LBX	rs rt rd	$\{24\{M[R[rs] + R[rd]](7:0)[7], M[R[rs] + R[rd]](7:0)\}$
010011	LBUX	rs rt rd	$R[rt] = \{24'b0, M[R[rs] + R[rd]](7:0)\}$
010100	LWX	rs rt rd	$R[rt] = M[R[rs] + R[rd]]$
010101	MUL	rs rt rd	$R[rd] = \{R[rs] \times R[rt]\}(31:0)$
010110	MULH	rs rt rd	$R[rd] = \{R[rs] \times R[rt]\}(63:32)$
010111	MULHU	rs rt rd	$R[rd] = \{R[rs] \times R[rt]\}(63:32)^{(8)}$
011000	DIV	rs rt rd	$R[rd] = R[rs] / R[rt]$
011001	DIVU	rs rt rd	$R[rd] = R[rs] / R[rt]^{(8)}$
011010	REST	rs rt rd	$R[rd] = R[rs] \% R[rt]$
011011	RESTU	rs rt rd	$R[rd] = R[rs] \% R[rt]^{(8)}$
011100	ADD	rs rt rd	$R[rd] = R[rs] + R[rt]$
011101	SUB	rs rt rd	$R[rd] = R[rs] - R[rt]$
100000	TRAP	aux	$EPC = PC + 4; PC = M[\text{aux} \ll 2]$
100001	RFT		$PC = EPC$

Tabla A.2: R-type instructions sorted by funct